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"Quantum 3.0": What Will It Mean for War, Peace, and World Order?

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Without quantum mechanics, the nuclear and information revolutions of the twentieth century would not have happened. In this article, we examine how new quantum technologies are coming to reshape war and peace in the twenty-first century. Research and development across quantum computing, communications, and sensing has taken on the dynamic of a "quantum race" between major powers, big tech, and academic institutions. A quantum advantage in these areas could radically reshape political and economic as well as strategic balances of power. In a world already marked by global entanglement, deep uncertainty, and superpositional states of war and peace, quantized modes of understanding need to expand from the natural and physical sciences into the social and political sciences, which remain stuck in causal, linear, and deterministic worldviews inherited from the seventeenth century. We use this article to begin to push beyond the classical limits of traditional international relations to consider new epistemic tools, ethical frameworks, and geopolitical strategies for a world order transformed by Quantum 3.0.

"EVERYTHING IN THE PAST IS A PARTICLE..."

Without quantum mechanics, there would be no atom bomb or nuclear revolution, no microprocessor or information revolution. Each of these revolutions triggered shifts in scientific paradigms, altered social realities, and reconfigured global power. In the first quantum revolution, thermonuclear weapons helped end a world war, start a cold war, and dominate a bipolar order; in the second, information technology helped network societies, hyperglobalize the economy, and balkanize global politics.

If measured by the catalysts of these revolutions—the invention of the atomic bomb, semiconductor, computer, laser, optical fiber, GPS, and smartphone—quantum mechanics is not only the most scientifically successful but also the most technologically transformative theory in modern physics (Orzel 2015). Like the harnessing of thermodynamics that produced the external (steam) and internal (gas) combustion engines, or of electrodynamics that led to electrification, the telegraph, and radio, the quantization of physics and increasingly other scientific fields has been characterized by a sequence of theoretical discoveries, experimental proofs, technological breakthroughs, economic disruptions, and geostrategic applications. In the

process, enormous wealth has also been produced: at the turn of the twentieth century, it is estimated that already 30 percent of the gross national product of advanced economies came from innovations made possible by quantum mechanics (Folger and Gall 2001; Grumbling and Horowitz 2019).

A third quantum transformation is on the horizon. Incubating in governmental, university, and corporate laboratories around the world, and still too early to call a revolution, "Quantum 3.0" rivals prior innovations derived from quantum mechanics, and includes new applications in computing, communications, sensors, control, simulation, and artificial intelligence. The building blocks of Quantum 3.0 are quantum bits or "qubits," which, entangled in probabilistic states of superposition (from 0 to 1), have demonstrated exponential advantages in speed, scale, and precision over classical systems based on digital bits in binary states (either 0 or 1). Quantum computing is progressing along several different pathways, including superconducting, trapped ion, silicon dot, photonic, and, most recently, topological. In October 2019 Google was the first to claim experimental realization of "quantum supremacy" (solving problems a classical computer cannot) when its 53-qubit Syracuse processor calculated in 200 seconds a complex

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circuit problem that would take a classical supercomputer 10,000 years to accomplish. Shortly after, IBM challenged Google’s claims and asserted that its 433-qubit Osprey chip would be a paradigmatic breakthrough (Arute et al. 2019; Pednault et al. 2019; Wehden, Faro, and Gambetta 2021). A year later, China’s University of Science and Technology claimed the holy grail of “quantum advantage” (solving real-world problems faster than a classical computer) with its 66-qubit Zuchongzhi computer, performing calculations in 20 seconds that would take a supercomputer 600 million years (Zhong et al. 2020). And most recently, Microsoft announced the development of a fault-tolerant, topological qubit for its Azure quantum computer (Langston 2022).

Quantum 3.0 is spreading into other critical strategic areas. Leveraging the unique random properties of quantum states, governments, banks, and other institutions are using quantum key distribution (QKD) to create secure communication networks. With China’s launch of the Micius satellite in 2016 and a subsequent space-ground link to Europe in 2020, the first steps have been taken toward building an extraterrestrial quantum internet. Quantum control at the subatomic level, already operating in technologies like magnetic resonance imaging (MRI), is advancing rapidly in the development of quantum sensors. Super-sensitive to changes in gravitational and electromagnetic fields, quantum sensors on land and in sea, air, and space have multiple strategic applications, including more precise and resilient global positioning, detection apparatus, and targeting systems. Quantum artificial intelligence, often described as “AI on steroids,” will be able to surpass the limitations of Moore’s law, in which transistors on a single chip will no longer be able to double every year as they approach the thinness of a single nanometer and begin to leak electrons because of quantum tunneling. The parallel processing power of qubits will in turn make quantum computers qualitatively superior to classical computers in adaptive machine learning, an essential requirement for future breakthroughs in neuromorphic computing and general artificial intelligence (Selman 2022).

“EVERYTHING IN THE FUTURE IS A WAVE...”¹

How will these large investments play out in the near and long term? As Niels Bohr wryly noted, predictions are difficult, especially of the future. The path to a quantum future is particularly uncertain, uneven, and multivariable. The number of players is limited by high costs. Supercooling, shielded facilities, rare materials, and intellectual capital are formidable entry barriers. Progress is impinged on by the inherently unstable nature of the qubit and high er-

ror rates. The price per qubit, currently hovering around \$8,000, will need to substantially drop to be commercially viable. A history of overhyped technologies, like AI, cold fusion, and virtual reality, invites skepticism. And above and beyond the scientific drivers, the human factor remains paramount and indeterminate: how will a world largely unschooled in the ways of quantum respond to an incredibly powerful and highly exotic technology?

Given the levels of global investments and technological innovations, there is a growing consensus among experts that within a decade, cloud-based universal quantum computers leveraging the special properties of qubits and quantum algorithms will have the capacity to produce unbreakable encryption, new classes of drugs and fertilizers, more efficient solar power and batteries, and better models for complex systems. Quantum information and material sciences, analyzing vast datasets, identifying informational inefficiencies, and generating dynamic routing solutions, will be able to optimize the flow of goods, resources, and finances beyond the capacity of any classical computational system.

On the darker side, quantum computers could also break existing public key cryptography based on factoring primary numbers and take surveillance data mining and face recognition to Orwellian levels of omniscience. Quantum command and control systems incorporating quantum sensors could provide superior analysis and presentation of situational awareness, boost logistical planning and operations, and simulate military scenarios at a level of scale, accuracy, and speed unimaginable by current standards. Quantum artificial intelligence could prove superior to human intelligence not only in pattern recognition but also in complex decision-making—and possibly learn to know it. Quantum augmented realities would pose a clear threat to privacy, as more of our personal lives and the public sphere is captured, encoded, analyzed, and utilized by opaque technological systems. The integrity of public discourse, based on a shared sense of reality and mutual understanding, would be further fragmented. In other words, and as is the case with almost every new powerful technology, Quantum 3.0 entails peril as well as promise.

Obviously, who reaches the next stage of quantum innovation first could introduce new and destabilizing asymmetries into the international order (Krelin 2021, 27). A hard look at the future implications of Quantum 3.0 is clearly needed. However, critical investigations within the social sciences have been handicapped by the persistence of worldviews largely inspired by Descartes, Hobbes, and Newton that date back to the seventeenth century.² As pointed out by Stephen Del Rosso, director of the Peace

1 “Everything in the past is a particle, everything in the future is a wave,” a quote attributed to Sir William Laurence Bragg (Nobel Prize in Physics, 1915), captures Niels Bohr’s complementarity principle, in which more than one truth about reality can be true, such as light can be both a wave and a particle until an observation or a measurement is taken.

2 Recent efforts to correct this intellectual aporia include Der Derian and Wendt 2020, 402; 2022; Fierke 2022; Murphy 2021; Wendt 2015; Zanotti 2018. See also Katzenstein 2022 for an extensive investigation into the analytical failure of traditional IR to manage uncertainty because of the persistence of outmoded Newtonian worldviews.

and Security Program at the Carnegie Corporation of New York, "Classical, Newtonian conceptions of the world, however refuted and transcended by quantum physics, provide a simple and comforting, if also inadequate, refuge from the messiness and indeterminacy of contemporary affairs" (Del Rosso, foreword to Der Derian and Wendt 2022, xiii). To be sure, sovereign states continue to behave as rational monads, go to war over territorial disputes, and bang up against each other like billiard balls. But historically they do so less often—which is one reason (among others provided below) why the Ukraine-Russia war has proven to be such a shock to the international system.

Much more prominent is a series of contemporary events that have evaded the anticipatory as well as explanatory powers of traditional international relations theories. Built from the ruins of the Second World War, enshrined with democratic values of freedom, equality, and peace, and standing in opposition to the closed systems of the Eastern bloc, Liberalism emerged from the end of the Cold War as the dominant discourse of the international order. The concerted globalization of the production, exchange, and consumption of goods and services, exemplified by the establishment of the World Trade Organization in 1995, was, according to President Bill Clinton, now going to bring "growth at home through growth abroad," and American strategic policy was aligned with the Liberal conception that the nation could counter "threats to [their] interests by promoting democracy, economic growth and free markets, human dignity, and the peaceful resolution of conflict" (Clinton, quoted in Bacevich 2009, 86; Aspin 1993, 1). Joseph Nye and other Liberal scholars articulated a vision of an American-led international system that was to deliver peace and plenty through a system of soft power and economic interdependence (Nye 1990).

The end of history, intellectually celebrated and politically promulgated by an American-led Liberal order, proved short-lived. Following the 9/11 attacks, the Afghanistan and Iraq interventions resulted in highly costly wars rather than a democratic peace. Realism might have proven more prescient than Liberalism about the consequences of disproportionate wars in response to the 9/11 terrorist attack, but its state-centric worldview handicapped its ability to comprehend the asymmetrical effects of terrorism and the dynamics between state and nonstate actors that proved so important during this period. Nor could Realism, relying on a rationalist and materialist conception of power, understand the observational powers of a global media that effectively force-multiplied a \$500,000 terrorist attack into a \$3 trillion (and counting) "Long War."

Geopolitical uncertainty and economic volatility persist, and the international order is now beset by the highest number of major conflicts in recent history (fifty-six) and the lowest number of democracies (thirty-two).³ The center

no longer holds: the United States, divided at home and facing a rising economic and strategic peer competitor abroad, is barely able to live up to its name, let alone retain the self-assigned title of leader of a "rules-based" international order. Liberal policies have been supplanted by programs of economic decoupling, supply chain "derisking," and the formation of new security blocs (like the Quadri-lateral Security Dialogue, or the Quad, and the Australia-United Kingdom-United States trilateral security pact, or AUKUS). Much as the peaceful dismantling of the Soviet Union and the American victory in the Cold War led to a drive to develop more nuanced, and indeed more speculative, theories of international relations, so, too, do present global dynamics require new conceptual and theoretical tools that are better equipped to account for systemic complexity and uncertainty.

During this interregnum, Quantum 3.0 could emerge not only as the source of more powerful technologies but also as a useful geopolitical heuristic. We also acknowledge, however, that it faces not only major scientific challenges but also considerable intellectual resistance. The counter-intuitive and mathematical formalism of quantum mechanics has a history of deterring acceptance both within and outside the scientific community. From its beginnings, quantum mechanics proved a particularly paradoxical field of knowledge. Niels Bohr established an early litmus test, declaring that "those who are not shocked when they first come across quantum theory cannot possibly have understood it." This view would be echoed by other eminent quantum physicists, including Roger Penrose ("quantum mechanics makes absolutely no sense"); Richard Feynman ("it is safe to say that nobody understands quantum mechanics"); and Freeman Dyson ("classical means that things are ordinary, quantum means that things are weird").

More than an immanent complexity is at work. Critical investigations face obdurate multidisciplinary challenges, from physical scientists who insist that quantum phenomena cannot scale up from the subatomic to the macrophysical level; from social scientists who belittle quantum concepts as mere metaphors; and from philosophers who view the effort to import quantum theory as one more instance of "science envy." For reasons of space, we can only offer here a shortened version of responses provided in greater detail elsewhere (see Der Derian and Wendt 2022).

First, microphysical quantum phenomena at the molecular and atomic levels have already been shown to scale up to biological, cognitive, and behavioral levels, including in photosynthesis, in bird migration, and, indirectly, through decision theory (Busemeyer and Bruza 2012; Al-Khalili and McFadden 2014). Second, metaphors matter, especially those whose metaphorical nature has been forgotten (pace Nietzsche) and elevated to the status of a natural law, like the classical metaphor of sovereign states as black-box, ki-

3 See Herre 2022. There have also recently been some notable swings in these figures—which is the point. A quantum perspective implies uncertainty, up until a point of observation/measurement, which inevitably entails a political interpretation. A quantum perspective also helps us avoid the binary of a tragic realism or a naive idealism as the only possible responses to such figures. See Tett 2023.

netic objects functioning in a clockwork balance of power. Third, *how* metaphors matter has shifted phenomenologically, in the sense that global events are increasingly *experienced* through densely networked media as quantum phenomena. When former US secretary of state George Shultz and arms negotiator (and theoretical physicist) Sidney Drell invoked the appearance of a new “quantum diplomacy,” or former president of Armenia Armen Sarkissian (also a theoretical physicist) espoused the idea of a globalizing “quantum politics,” they also intimated the need for a new phenomenology to understand quantum-like events and behavior.⁴

All kinds of odd and morbid symptoms appear when an outdated classical worldview comes up hard against new quantum realities. From the fall of the Berlin Wall, to the toppling of the Twin Towers, to the financial crash of 2008, to the Black Lives Matter movement, to the Ukraine-Russia war, the late twentieth and early twenty-first century seem etched by quantum-like phenomena that have been catalyzed and globalized through networked media. Just as radiation can “flip” bits in computers or coding in a strand of DNA, these *single-event upsets* can produce positive changes as well as negative mutations. What has been lacking in almost every instance is a heuristic that can help anticipate or enhance the positive and prevent or mitigate (rather than worsen) the negative effects of such flips.

While Realists and Liberals might differ on the future of the world order, they share a rearward-facing, state-centric, classical worldview, bound by an imaginary of a return to the multipolarity of the nineteenth century, the bipolarity of the American and Soviet empires, or, not surprisingly, the unipolarity of Pax Americana at the end of the Cold War. Our advocacy for a new intellectual response goes beyond an assessment of past and anticipation of future global events. We believe the convergence of quantum science, quantum tech, and increased quantum-like phenomena could entail a tipping point in an ongoing transformation of world order, from a classically bounded state-centric system to a global *heteropolarity* in which big-tech corporations, nongovernmental actor-networks, and superempowered individuals present their own alternatives to classical statecraft.

QUANTUM TECHNOLOGY IN WAR AND PEACE

Some history is hard to escape: every powerful new technology is eventually weaponized, and quantum is unlikely to be an exception. AI has already been incorporated into surveillance, targeting, and decision-making in warfare. Cyberwar, infowar, and other forms of digitized influence operations are now essential tools of just about every advanced state. However, on the other side of the war-peace spectrum, cyber diplomacy, social media, and open-source intelligence have become new forces for mobilizing civil so-

ciety and coordinating governmental and nongovernmental efforts to counter coercive and antidemocratic regimes.

The early history of quantum mechanics is similarly bifurcated by aspirations of peace and the necessities of war. Many of the early conceptual advances came from an interwar moment of international collaboration. Moreover, the pioneering work of theoretical physicists like Max Planck, Albert Einstein, Niels Bohr, Werner Heisenberg, and others influenced thinkers in fields well beyond the physical sciences, including philosophers and political scientists, to incorporate a more complex understanding of physical reality into their worldviews (Der Derian and Wendt 2020, 402). However, the rise of fascism and the urgency to build the first atomic bomb nationalized and weaponized quantum mechanics. And while many of the current advances in quantum computing, quantum communications, quantum sensing, and the production of quantum materials largely resulted from high-level, international and collaborative research, the potential to dramatically enhance military capabilities, disrupt established strategic practices, and make considerable profits in the process once again threatens to return quantum research into national and private corporate silos.

The key element of the current race lies in the incorporation of the probabilistic mechanisms of quantum physics into information processing. In the security world, this is likely to lead to advances in a number of directions. Quantum encryption/decryption, internet, sensors, and GPS are already being funded and developed for security and military use around the world; who gets there first is likely to produce new asymmetries of power. It could result in a leap in AI and machine learning, which would improve the performance of lethal autonomous weapons systems in both their selection and their engagement of targets, as well as vastly increased capacity for analyzing large datasets used in defense intelligence and cybersecurity. It will also confer a major security advantage in the realm of cyberattacks to both civilian and military infrastructure. A quantum attack would compromise the security of the entire network of a rival’s digital commerce and communications if it remained operating on classical encryption standards. This has spurred research into post-quantum cryptography (PQC) methods for protecting data that are less vulnerable to quantum computer hacking. The US National Institute of Standards and Technology is now in the fifth year of a public submissions process for evaluating quantum-resistant encryption algorithms upon which new public-key cryptographic methods will be based, and has developed guidelines for the migration of classical encryption systems to PQC protocols (Murugiah and Barker 2021).

Although there has been a consistent increase in research funding into quantum computing in recent years, it remains in the nascent stages of development. Despite claims from private companies of having crossed the

4 See Shultz 1998; Der Derian 2013, 2022; and Sarkissian 2020. The founder of the World Economic Forum, Klaus Schwab, also applies the idea of quantum politics in Schwab and Malleret 2020.

threshold into “quantum supremacy,” of solving problems that a conventional computer cannot, a universal quantum computer is unlikely to *replace* classical computers (Grumblin and Horowitz 2019, 157). The impact of quantum computing on international security will be constrained by the capacity to build networks that rival the resilience and redundancy of the current internet. The foundational research for the development of quantum networks is underway, with the United States and China exploring both land- and space-based quantum entanglement distribution. This would allow quantum computers to interlink with quantum sensing and communications capabilities, enhancing the ability to process data gained through sensors and in the decryption of intercepted sensitive material.

Quantum communications technology also holds significant implications for international security, and the research and development of quantum communications networks is already well advanced. The major focus of quantum communications is in the transport and exchange of information by encoding it into a quantum system, a necessary capability for the development of networked quantum computing (Grumblin and Horowitz 2019, 29). A subfield of quantum communications, quantum cryptography, also allows for the development of communications systems that are impervious to interception by unwanted observers. This is an area of quantum research in which the Chinese government is a clear leader. It launched its first Quantum Experiments at Space Scale (QUESS) satellite in 2016 and completed its first quantum key distribution (QKD) optical network between Beijing and Shanghai in 2017. In January 2021, China announced the completion of a 4,600-kilometer integrated quantum communications network capable of video, audio, text, and file transmission across much of the country, using a combination of quantum satellites and optical cables (*Xinhua News* 2021). The United States has enjoyed a clear advantage in intelligence gathering, communications, and surveillance for the past seventy years. Breakthroughs in the field of quantum communications could overturn—and, if coupled with advances in quantum computing, potentially invert—these advantages.

Quantum sensing is the area of quantum technology that is most ready for military application today. By utilizing the properties of quantum particles, such as the entanglement of photons, new sensing systems are able to much more accurately detect and measure the physical environment, including distortions in magnetic and electric fields, gravity, and temperature. This has implications for the resilience and accuracy of GPS systems, radar, and the detection of stealth weapons, including the latest generation of assault aircraft and nuclear submarines.

The maintenance of a reliable second-strike capability is a major pillar of the underlying logic of mutually assured destruction. Should a nuclear-armed power suffer an attack by nuclear weapons, even to such an extent that their capacity to launch a retaliatory attack from their land-based nuclear arsenals is compromised, its nuclear submarine force ensures that any potential foe would suffer catastrophic nuclear retaliation. The impact that quantum sensing could have on submarine survivability is being

taken seriously by the US government (Sayler 2022, 1). The furthest developed applied quantum sensor in this domain is the superconducting quantum interference device (or SQUID), which detects subtle electromagnetic disturbances. These have been in use since the early 1960s in biological sciences and medicine, but SQUIDs are now being developed as airborne devices to detect submarines. Conventional airborne detection devices are effective only in the range of a few hundred meters, while SQUID-based magnetometers could detect submarines at an estimated range of six kilometers or further. This is a major improvement in sensitivity and range, but experts maintain that quantum sensing capabilities that could truly make the oceans transparent, and endanger the near-invulnerability of submarine-based nuclear forces, will not occur in the short to medium term (Kubiak 2020). Still, the potential for quantum sensing to confer such an advantage as to degrade the stealth capabilities of nuclear submarines could force a recalibration of strategic nuclear forces and their deterrent effect in the years to come.

Advances in quantum sensing also plays into a kind of “budgetary war” in the sense that developments could, for example, make much of the most expensive project in the history of the US armed forces, the Joint Strike Fighter, obsolete. China has already successfully engaged in a type of asymmetric budgetary warfare against the United States through its strategic focus on Anti-Access/Area-Denial (A2AD) capabilities in the Western Pacific. American power projection capabilities in the region center around its five Pacific carrier strike groups and vast network of military bases, all incredibly costly to build and maintain and complex to operate. China’s focus on developing missile defense, most notoriously the DF-21D “carrier killer” anti-ship ballistic missile, has given it a clear budgetary as well as strategic advantage, in what can be interpreted as an inversion of the arms race dynamic that saw the United States escalate Soviet spending to unsustainable levels toward the end of the Cold War. A 2016 RAND study commissioned by the Department of Defense found that, due to China’s development of A2AD military assets, the United States could no longer be confident of winning a war in the Western Pacific (Gompert, Cevallos, and Garafola 2016, iv–v). Developments in quantum technology could render a similar asymmetric advantage on a grand scale.

The development of these kinds of quantum capabilities is not discrete, and quantum advances in other areas could also improve the performance of stealth capabilities and concealment—for example, through quantum navigation and quantum stealth materials technologies (Chen et al. 2019; Feng 2019). However, the capacity of quantum technology to degrade the performance capabilities of state-of-the-art weapons platforms and to disrupt stable pillars of strategic nuclear deterrence are instructive examples of its potential to reshape the strategic landscape in years ahead.

A final area of quantum research that has the potential for significant strategic and geopolitical disruption is the study of quantum materials. These are nanomaterials that exhibit unique quantum mechanical properties like quantum entanglement and quantum fluctuation, and that are



Figure 1. Technology readiness levels (TRLs) are a method for estimating the maturity of technologies during the acquisition phase of a program used by the Department of Defense, which allows for standardized measurement and comparison across different technologies. Quantum computing is currently considered to be at TRL 5, at which point error correction is the primary obstacle to be overcome.

Source: US Department of Defense, Defense Acquisition Guidebook (2013), 848, and Kristel Michielsen and Thomas Lippert, Forschungszentrum Jülich, “Quantum Technology Readiness Levels Describing the Maturity of Quantum Computing Technology.”

constructed in such a way as to create conditions for and exploit superconductivity and other electromagnetic properties. These materials will be used for the production of the aforementioned quantum technologies, and they are also envisioned to play a role in the recently inaugurated project to construct critical technology supply chains for the United States that are independent of Chinese inputs. The use of quantum materials, particularly topological insulators, is being investigated in the production of new and powerful semiconductors by the US Defense Advanced Research Projects Agency (DARPA) in collaboration with the broader scientific community. This research contributes to a US\$1.5 billion DARPA megaproject, the Electronics Resurgence Initiative, which seeks to investigate new materials and chip designs to eliminate dependence on Chinese sources of critical materials in the American technology sector (Atlantic Council 2021, 54).

Carbon nanomaterials with quantum properties have been demonstrated to function as substitutes for what were previously considered to be irreplaceable critical minerals (Arvidsson and Sandén 2017). While research into substituting quantum materials is still in very early stages, and clear and significant barriers exist today in terms of scalability, energy use, and ancillary environmental impacts, the potential for the substitution of critical materials has enormous implications for geopolitics. Researchers from the United States Geological Survey (USGS) have found that the United States and China have eleven minerals in common for which they are import reliant for at least 50 percent of their supplies (Gulley, Nassar, and Xun 2018, 4111). China is also the single country upon which the United States is most reliant, for twenty-five separate minerals that it has a greater than 50 percent import reliance (US Geological Survey 2019). An African or a Latin American country is one of the two primary sources of supply for each of the eleven

minerals identified by the USGS scholars as being shared as critically import reliant by both China and the United States (Gulley, Nassar, and Xun 2018, 4113). Security of supply of strategic raw materials has been a major factor in security competition between states since at least the First World War. The United States and China are clearly entangled in terms of the supply of critical raw materials, and should economic decoupling and the shoring up of independent supply chains continue, competition for these critical materials could be fierce. The development of quantum materials as a substitute for these critical minerals could greatly mitigate the risk of resource conflict.

THE QUANTUM WORLD ORDER: UNI-, BI-, OR HETEROPOLAR?

The quantum race between China and the United States, and how it will influence the global balance of power, has been viewed largely through a bipolar prism. Close attention to the US-China quantum rivalry is certainly justified: they are the two clear leaders among states in terms of quantum patents and the production of working quantum technology, are two of the biggest state investors into quantum research, and are host to the largest and most successful private enterprises in the quantum space. There is every indication that the quantum race, paired with the ongoing strategic, economic, and broader technological competition between the two countries, heralds a new era of superpower rivalry. This could well lead to a return to a bipolar Cold War-style system, albeit with some clear ideological and material differences between it and that of the latter half of the twentieth century, that sees the United States and China competing for global domination under new technological conditions. The much-needed focus on US-China power dynamics and quantum should not, how-

ever, blind us to the potentialities of the quantum worldview and heteropolarity, discussed in the introduction to this piece. It is important to remember that at the origins of the digital revolution, almost every governmental and corporate leader thought that the world would never have—or need to have—more than a dozen mainframe computers.

The embedded costs of developing quantum technology do currently ensure that only a relatively small number of actors are capable of competing in the field. Large amounts of capital investment, supercooling facilities shielded from even minuscule levels of ambient environmental interference, critical raw materials, and intellectual capital in the form of mastery of quantum theory, as well as scientific and engineering expertise, all form a remarkably high barrier to entry in the quantum race. The development of more complex technologies with military applications, dependent on the mustering of resources and technical abilities possessed by fewer actors, concentrates the production of weaponry in advanced states and leads to their ability to exercise power over people, states, and regions of the world who do not enjoy the same level of techno-industrial development. Quantum technology is likely to exacerbate this inequality of capacity and power between states, and, as is to be expected, the world’s most powerful states have already established an imposing lead in the field.

The US government has been rapidly expanding its organizational capacity and investment into quantum technology in recent years. Federal quantum R & D has grown from US\$450 million in 2019 to US\$710 million in 2021 (Subcommittee on Quantum Information Science 2021, v). Much of this funding flows to private and university research projects, but there has also been a proliferation of in-house federal agency research into quantum technology. Under the “Three Pillars Model” established by the federal government, funding to create and sustain an American quantum ecosystem is split between civilian, defense, and intelligence agencies. Quantum research is one of the Department of Defense’s top modernization priorities. Research coordination offices and funding streams have been established across the US Army, Navy, and Air Force, as well as through DARPA (Subcommittee on Quantum Information Science 2021, 15). The American intelligence community has also been a major actor in quantum research and development. The National Security Agency has funded quantum research through its Laboratory for Physical Sciences since the 1990s. The CIA conducts its own quantum research and development in collaboration with academia and the private sector through the newly minted CIA Labs, and also directs strategic investment into quantum research through its In-Q-Tel fund. The Intelligence Advanced Research Projects Activity (IARPA), a cross-agency research collaboration that sits under the Director of National Intelligence, lists quantum as one of its four primary research areas, along with AI, machine learning, and synthetic biology, and both conducts and supports a wide range of quantum research, with a focus on quantum computing.

Despite the major increase in federal investment into quantum R & D, the United States is actually slipping comparative to other nations. The United States is still the

leading private market for quantum technology, with over US\$2.3 billion raised in private markets in both 2021 and 2022, a major increase over previous years that accounts for around 68 percent of all private investment in the sector (Bogobowicz et al. 2023). But in quantum, the private sector is dwarfed by the state. Sovereign investment in the quantum sector has truly exploded in recent years. Recent industry estimates show that between 2015 and 2023, China’s investment into quantum technology grew from US\$200 million to US\$15.3 billion, a rate of almost 8,000 percent in eight years (Mahendran et al. 2021, 9). In these years, the United States has lost its lead. In 2015 it was the world’s largest investor in quantum, totaling almost US\$350 million; by 2023 this investment had expanded to US\$3.7 billion, but the United States had slipped behind both China and the European Union (US\$8.4 billion) in terms of state spending on quantum (Bogobowicz et al. 2023).

Chinese leadership is keenly aware of the decisive role that technology will play in the twenty-first-century struggle for strategic and economic power. The need for military modernization and technological development was prominently identified in China’s most recent defense white paper, released in 2015 (China 2015). The following year, China released a National Innovation-Driven Development Strategy, which focused on the development of quantum, among other new technologies, as critical “to achieve the Chinese dream of the great rejuvenation of the Chinese nation” (Central Committee 2016). President Xi Jinping has identified quantum science and technology as the frontier field in a new round of sci-tech and industrial revolutions, and stressed its role in accelerating the development of China’s economy and national security (*Xinhua* 2020). As a relative latecomer to the field, the Chinese government has invested heavily in its project to become the global leader in quantum science. As in the United States, quantum developments in China are deeply embedded in the national security establishment. While specific information on both funding amounts and research projects is difficult to ascertain, the People’s Liberation Army is known to be supporting research into quantum technology, through the National Defence Key Laboratories Fund, the Academy of Military Science, and the National University of Defence Technology (Kania 2021, 936).

The strategic elements of US-China quantum rivalry are clear, but quantum also holds the potential to greatly influence global economic leadership. The United States has enjoyed a preeminent position in technological innovation since the Second World War. American economic leadership has formed the basis of American world power, equipping it with immense influence over the economic destinies of much of the rest of the world, and allowing for the colossal outlays in military spending required to maintain the world’s most powerful and advanced military machine. In recent decades, China’s rise and the growing importance of digital technology to the global economy has shaped a new economic geography dominated by the United States and China. These two countries account for 90 percent of the market capitalization of the world’s largest digital platforms; Europe’s share is 4 percent, and Africa and Latin

America combined account for only 1 percent (UNCTAD 2019, xvi).

The prominence of digital technology in the American and Chinese economies—and the much higher growth rate of the digital compared to the traditional economy in recent years—means that supremacy in the digital sphere will be the major determinant of relative economic power between the two countries over the coming period (Xuetong 2020, 323–24). William Burns, director of the CIA under President Biden, has stated that technological competition will be the main arena for rivalry with China in the years ahead (Burns 2021). China has already come to supplant the United States in several areas of global economic leadership. Over the past forty years, China has transitioned from a marginal economic outsider to the world’s largest economy (by purchasing power parity), trader, manufacturer, and holder of foreign exchange reserves (Morrison 2019). China is the largest export destination for thirty-three countries and the largest source of imports for sixty-five (Woetzel 2019, 2). It is the biggest trading partner of the United States and the largest single holder of US treasury bonds. It is also the largest trading partner of most countries in East Asia, including all of the major economies and staunchest American allies.

The United States still holds a significant, but narrowing, lead in the key industries of the twenty-first-century economy, particularly digital networked communications, aerospace engineering, arms manufacturing, pharmaceuticals, and green technology, all areas in which quantum holds the potential for major advancements. Securing the American “National Security Innovation Base” and “Defense Industrial Base” has become a top priority in strategic planning circles in the United States. In a near inversion of the liberal theory of economic integration leading to international peace, the new dominant approach to US-Chinese economic relations is that China is a threat to national security, due largely to the deleterious effects of technology transfer, open and illicit, on America’s competitive edge. A Department of Defense strategic planning report describes the situation as such: “Although trade has benefitted both China and its trade partners, Chinese use of espionage and theft for economic advantage, as well as diversion of acquired technology to the military, remains a significant source of economic and national security risk to all of China’s trading partners” (US Department of Defense 2019, 9).

Perhaps the most comprehensive piece of American policy addressing the emerging threat of Chinese techno-industrial competition has been the “Clean Network” program. This was established by the Trump administration to exclude “untrusted” Chinese carriers from the US telecommunications network, remove Chinese applications from American mobile app stores, exclude Chinese businesses from accessing personal and proprietary information held in American cloud storage, and ensure the integrity of the physical infrastructure, like undersea cables, that underlie digital networks (Pompeo 2020). The Clean Network program explicitly targets China alone, and encourages other countries to do the same by incentivizing their own access

to America’s “clean” network of government and commercial digital communications. Over thirty other countries, and many major telecommunications companies, have already signed on to the program (Pompeo 2020). It is believed that locking China out of the digital economic and telecommunications networks that provide it with access to US-developed intellectual property and proprietary information will critically weaken China’s capacity to compete in the high-tech economy, hamper its ability to transition to a fully modern military, and ultimately loosen the Chinese Communist Party’s grip on power (Trump 2017, 21, 29–32).

The same approach to technological competition with China has predominated under the Biden administration. In an address to a joint session of Congress, Biden stated that to win the twenty-first-century struggle against China, “We have to develop and dominate the products and technologies of the future....” (Biden 2021). In the first trip to Europe of his presidency, Biden managed to put a joint G7 international infrastructure financing and development program to rival China’s Belt and Road Initiative (BRI) on the agenda—the Build Back Better World (B3W) partnership—and left a NATO conference with a joint communiqué that frames China as a systemic challenge to the international order, and a threat to the security of the NATO states, for the first time ever (White House 2021; NATO 2021).

In October 2022, the Biden administration announced a set of sweeping controls on the export of advanced-computing and semiconductor manufacturing items to China (US Bureau of Industry and Security 2022). Gregory C. Allen, director of the AI Governance Project at the Center for Strategic and International Studies, described it as the beginning of “a new U.S. policy of actively strangling large segments of the Chinese technology industry—strangling with an intent to kill” (Allen 2022). This export ban followed closely on the heels of the signing of the CHIPS and Science Act, which authorized \$52 billion of government incentives to support the US semiconductor industry across research and development, to secure supply chains, and to ensure the “reshoring” of manufacturing (White House 2022). Speaking in the weeks following the implementation of the semiconductor export ban, Secretary of Commerce Gina Raimondo identified quantum information systems as a critical technology that the US government would take action to protect through further export controls and investment screening regimes, to prevent China from eroding America’s competitive advantage (Raimondo 2022).

The first salvos in the battle for twenty-first-century techno-industrial and military supremacy have already been fired. The retreat of the world system toward one of superpower rivalry, distinct economic spheres, and intense strategic brinksmanship over the past decade is striking. UN secretary-general António Guterres has warned of “a new risk looming on the horizon.” He described the end-point of rising US-China tensions as entailing “the possibility of a great fracture: the world splitting in two, with the two largest economies on earth creating two separate and competing worlds, each with their own dominant currency, trade and financial rules, their own internet and artificial

intelligence capacities, and their own zero sum geopolitical and military strategies” (UN News 2019). Quantum is seen as a keystone technology that could deliver decisive advantage in this struggle in both the United States and China.

While the observable effects of quantum competition in international relations have pointed toward a new era of superpower rivalry thus far, there remains the possibility that this technology could be so revolutionary as to fundamentally reshape the underlying structure and interconnections of world order itself. The nature and the potential of quantum technology make it qualitatively different from other major technological-military breakthroughs of recent history. The case of nuclear weapons is often raised as a comparator, involving a race to develop a technology, already theoretically proven by physicists, which would give a major advantage in strategic affairs and divide the world between nuclear “haves” and “have-nots.” Conceptualizing quantum in this way is a mistake on multiple levels. Quantum actors have a clear asymmetrical potential in a way that the large state-actors of the nuclear weapons club do not. Private enterprise, with its complex, and sometimes conflictual, relationship with the state, plays a much larger role in Quantum 3.0 than in the period that produced the nuclear revolution or than the government (largely the military) that funded the infrastructure, from the transistor to the internet, of the digital revolution.

The race for quantum supremacy could, in fact, turn the logic of mutually assured destruction on its head. The advantages that quantum supremacy would confer are so great that there is a clear strategic logic behind conducting preemptive strikes should it appear that a rival is developing a clear and insurmountable quantum lead. Ian Bremmer has argued that this structural threat is so great that governments must immediately prioritize sharing information on developments in quantum computing, because “even the threat of such a breakthrough could trigger World War III....” (Bremmer 2022, 172). It is as if a portal is reopening, not unlike the one Oppenheimer and Bohr envisioned after they witnessed the devastation wrought from their theoretical knowledge of quantum mechanics, in which nation-states would be *forced* to cede some degree of sovereignty and to impose international controls to ensure the responsible use of nuclear energy.

There is also a great deal of potential in quantum cloud computing and the quantum internet, which could produce complex international and substate interactions, incentives, and relationships. One example of how this might work can be found in the exponential growth in computational power of networked quantum computers. The computational superiority of a quantum internet provides a clear material incentive for states and corporations to work together, networking their quantum assets and agreeing to best practices, regulations, and standards in order to share the mutual benefits.

Some adherents of game theory believe that this will result in cooperation and different points of leverage over access to the quantum internet between technologically advanced, powerful nations with a significant quantum capacity and weaker, poorer states who will need to provide

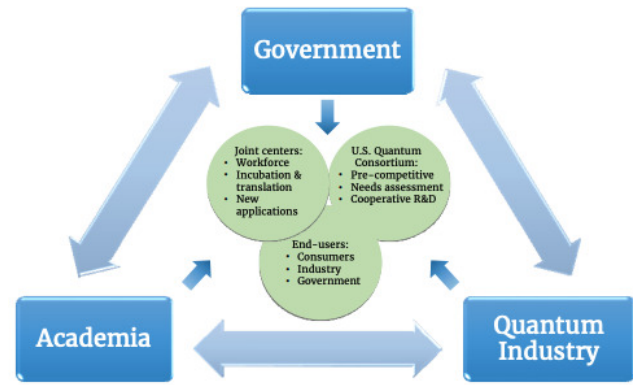


Figure 2.

Source: National Science and Technology Council. “National Strategic Overview for Quantum Information Science”. Edited by the Executive Office of the President of the United States, September 2018.

some other political value to be included in the quantum internet club (Rohde 2021, 316). Similar benefits to cooperation in international relations have been emphasized by Liberals for a long time. Faith that the benefits of quantum cooperation will be so large and dangers so grave that national rivalries will be circumvented is perhaps hoping for too much. But the incentive structures of quantum cooperation make it a more dynamic and unpredictable technological factor in world order than the stable deterrence of nuclear arms.

The research, production, and use of quantum technologies is occurring within a particularly interconnected and robust network of linkages between government, private industry, and universities. There is, of course, a long history of cooperation, coordination, and connected funding streams between these organizations, but their level of interdependence within the ecosystem of Quantum 3.0 suggests a more dynamic, less hierarchical relationship than has traditionally been the case. This implies that the exercise of power and the direction of research within this network is more complex than in the major military-industrial-research projects of earlier quantum revolutions, like the Manhattan Project. It also means that the boundaries delineating the constituents of the military-industrial-academic complex have become more opaque than ever in the quantum field.

In the case of the two leading national funders of quantum research, the United States and China, the direct funding of academic research by defense or intelligence organizations is nothing new; nor is the inordinate influence of what President Eisenhower first identified as the military-industrial complex. The Second World War ushered in what was described by MIT president James R. Killian Jr., the first science adviser to the office of the president, as a “golden era in government-university cooperation” (Abrams 1989, 22). The combination of novel and highly technical aspects of modern warfare, as well as America’s position as leader of the Western bloc during the Cold War, required the training of a large cohort of young scientists, engineers, bureaucrats, and managers to identify, deter, and if necessary defeat real as well as imagined threats. China’s rise has been

predicated on an even tighter fusion of university research, corporate development, and national security imperatives in high technologies, and with it, a mimetic rivalry with the West. Opinions differ on the extent to which economic competition, ideological difference, or power politics is driving this rivalry toward a new Cold War. Since Quantum 3.0 figures large in all three areas, it is highly likely to reshape the configuration of the world order. Whether its revolutionary potential will be realized as an open, internationalist *human science* or a closed, nationalist *military science* remains the key political question. We can take some solace from quantum by treating the "measurement problem" as an ethical imperative and a political opportunity to reinsert the power of individual choice in the determination of an uncertain quantum future.

IF NOT NOW, WHEN?

We have sketched out an incipient quantum worldview to help imagine what might come next in the long century of quantum mechanics, as a first step toward developing the necessary epistemic tools, ethical frameworks, and geopolitical strategies for a Quantum 3.0. In the absence of a tangible signifier, like the iconic image of the mushroom cloud or the thinking robot that marks the beginning (and possible end) of the first and second quantum revolutions, the quantum imaginary for a potential third revolution has suffered simultaneously from seeming too small, too cosmic, or just too strange to be taken seriously outside the fields of physics and other hard sciences. But as the next wave of quantum theory, science, and technology converges, we must develop a commensurable quantum worldview to anticipate and manage a world order reconfigured by the next wave of quantum innovation.

Ethical discussions of quantum extend beyond the more direct questions of which states and actors will be able to utilize quantum technology, for what ends, and with what implications for equity and security, and necessarily include normative considerations. Scholars of ethics and peacebuilding have already begun to explore the implications of quantum entanglement, relationality, and nonlinear causality on the undergirding logic of moral frameworks. Arguing that a quantum worldview allows a wider scope for reconciling subjectivity and free agency with structural questions of actors and states and their relations within a world system, they open up ethical horizons beyond the framing of subjects as monads programmed to pursue self-interest against others, the primary presupposition of both realism and economic liberalism (Wendt 2015, 28; Zanotti 2022, 361).

We might never fully understand quantum weirdness at the subatomic level, where particles can be in two places at once and correlate across space and time in unpredictable ways until an observation or a measurement is taken. Most physicists would say these counterintuitive properties decohere, collapse, or wash out at the classical level. Yet in a world increasingly marked by global entanglement, deep uncertainty, and superpositional states of war and peace, quantum not only begins to make sense; it can also help us make *better* sense of that world. Providing superior epis-

temics ("ways of knowing"), ontics ("ways of being"), and praxis ("ways of doing"), quantum could well prove to be not only the most holistic but also the most *realistic* theory of world politics.

We also believe that quantum can help us make a better world. If, as Australian-born Nobel laureate physicist Lawrence Bragg claimed, "everything in the past is a particle, everything in the future is a wave," then quantum—as in constructivist and other theories based on potentialities—implies an ethical imperative. When observations alter the reality observed, we need to make them meticulously and take them seriously, which is why we have been openly speculative, radically empirical, and, we hope, ethically responsible in our effort to actualize the benefits and highlight the risks of Quantum 3.0.

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There are no competing interests to state for this paper.

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